



Aung Min Khant

Mechatronics Engineer

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PROFESSIONAL SUMMARY

Mechatronics Engineering graduate with hands-on experience in PLC-based automation, embedded systems, and machine control, complemented by professional experience in industrial commissioning and technical documentation. Seeking an Equipment or Automation Engineering role where I can apply my knowledge of control systems, computer vision, and hardware integration to improve machine performance and production efficiency.

PROFESSIONAL EXPERIENCE

Freelance Commissioning Engineer | EATON Thailand (Contract Engagement) | Jun 2026 – Jul 2026

- Co-authoring Cx Level 2 and Level 3 commissioning scripts (MOS files and PTU unit scripts) for EATON Thailand as part of a contracted data center commissioning engagement.

Data Center Commissioning Engineer | Longmotive Construction (Thailand) Co., Ltd. | Nov 2025 – Apr 2026

- Inspection Management (Malaysia & Thailand): Conducted delivery inspections of cable trays, VRV AC units, EMSB, RMU, and prefab MVP modules — verifying serial numbers, unit counts, ratings, FAT, and FWT records; verified underground pipe installation alignment against engineering drawings; supported LOTO (Lock Out Tag Out) procedures on-site.
- Commissioning Scripts & Tracking: Authored, verified, and submitted Cx Level 1 and 2 mechanical commissioning and testing scripts to clients via Autodesk Construction Cloud; engineered a custom submittal tracker that eliminated documentation gaps and maintained documentation completeness with zero missing records.
- Digital Integration & Coordination: Optimized project data workflows by integrating Color-Level Tag data into CxPlanner and coordinating with NV5 consultants to resolve revisions, improve data consistency, and support accurate technical documentation.
- Intern Mentorship & Operational Scale: Scaled field operations by training and leading an intern team, driving the successful execution of 1,000+ delivery inspections with absolute data integrity and zero documentation errors.

ACADEMIC PROJECTS

On-Gantry Dual-Sided ASRS | Ongoing University Research (IEEE target publication)

- Developed a 4-axis automated storage and retrieval system integrating computer vision (OpenCV) for automated slot identification and positioning, achieving 1mm positioning accuracy.
- Designed a gantry-mounted temporary shelf enabling retrieval or storage of up to 4 loads per trip, reducing travel cycles and improving energy and time efficiency across dual-sided shelving.
- Cycle time: 3 minutes per load; multi-load architecture enables up to 4x throughput improvement per gantry trip versus single-load systems.

AI Indoor Localization System | Ongoing University Research (IEEE target publication)

- Developed a 7-node indoor localization system using multi-sensor fusion (LoRa, RSSI, compass, path constraints) and Google Colab AI processing, enabling precise zone-level identification across 4 transmitters.
- Improved positioning resolution from ambiguity between 2–3 transmitter zones to accurate single-node identification by combining unprocessed RSSI values, heading data, and path constraint logic.

Kinetic Energy Harvesting System | Final Year Senior Project (2024–2025)

- Developed a kinetic energy harvesting system converting industrial vibrations and pedestrian footfall into electrical energy, achieving approximately 4.6W average output (0.675A, 6.875V).
- Designed for footfall capture at intervals of up to 1 second, ensuring continuous energy capture from human foot traffic.

Pneumatic Shirt Folding System | Published (2024–2025)

- Designed electro-pneumatic automation system achieving an 8-second folding cycle using rack-and-pinion mechanism, timing pulley gear, pneumatic cylinders, 3D-printed joints, linear guides, and acrylic/EPE foam structure.
- Published as "Guidelines from Idea to Implementation: A Project-Based Learning Model for Education in the Age of AI" at EDTA Conference, Chiang Mai University.

TECHNICAL SKILLS

- **Automation & Control:** PLC Ladder Logic (Omron CX-Programmer), HMI Design, PID Control, Automation Scripting
- **Embedded Systems & IoT:** ESP32, Arduino, IoT Monitoring, Sensor Integration, LoRa
- **Computer Vision & AI:** OpenCV, Multi-Sensor Fusion, Google Colab AI, Path Constraints
- **Electronics & Hardware:** PCBA Testing, Circuit Troubleshooting, Power Electronics, Soldering
- **Design & Structural Analysis:** AutoCAD, SolidWorks, 3ds Max, Stress & Vibration Testing, Statics/Dynamics
- **RF & Data Collection:** RF Communication, Wireless Protocols (LoRa), Data Acquisition (DAQ)
- **Software & Logic:** Python, C++, Microsoft Visual Studio, MATLAB
- **Data Center Commissioning:** Cx Procedures (Level 1–3), Script Execution, Site Status Tracking

TOOLS & SOFTWARE

AutoCAD	SolidWorks	3ds Max	CxDesigner	CxProgrammer	CxPlanner
Python	C++	OpenCV	Microsoft Visual Studio	MATLAB	Processing
Arduino IDE	Microsoft Excel	Microsoft Word	Microsoft PPT	Adobe Acrobat	Autodesk Construction Cloud

EDUCATION

Bachelor of Mechatronics Engineering in Automation | Assumption University, Thailand | 2025 | GPA: 3.12 / 4.00

ACHIEVEMENT

1st Place – 3D Computer Aided Design of Innovative Car Competition, Assumption University (2025)

LANGUAGES

Myanmar: Native **English:** Professional Proficiency